

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER EXAMINATION

UCCN2243 INTERNETWORKING PRINCIPLES AND PRACTICES

THURSDAY, 29 SEPTEMBER 2022

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instruction to Candidates :

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in section B, marks will only be awarded for the FIRST question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

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Section A (Compulsory Questions)

- Q1. You are assigned the following network 200.1.152.0/21. Using variable-length subnet masking (VLSM), create subnets as required for the network shown in Figure 1 (assume IP subnet-zero). Ensure that you always utilize the next available network in your IP plan.
- Calculate** the network address, and subnet masks for all the networks (Area 1 to Area 6). Hint: The number of usable addresses in Area 6 corresponds to the number of gateway interfaces. (14 marks)
 - Determine** the gateway addresses on router R3. Assume the respective gateway addresses is the last usable address. (6 marks)
 - Determine** the summarized route of Area 1 to Area 5, with the longest possible network prefix. (5 marks)
- [Total : 25 marks]

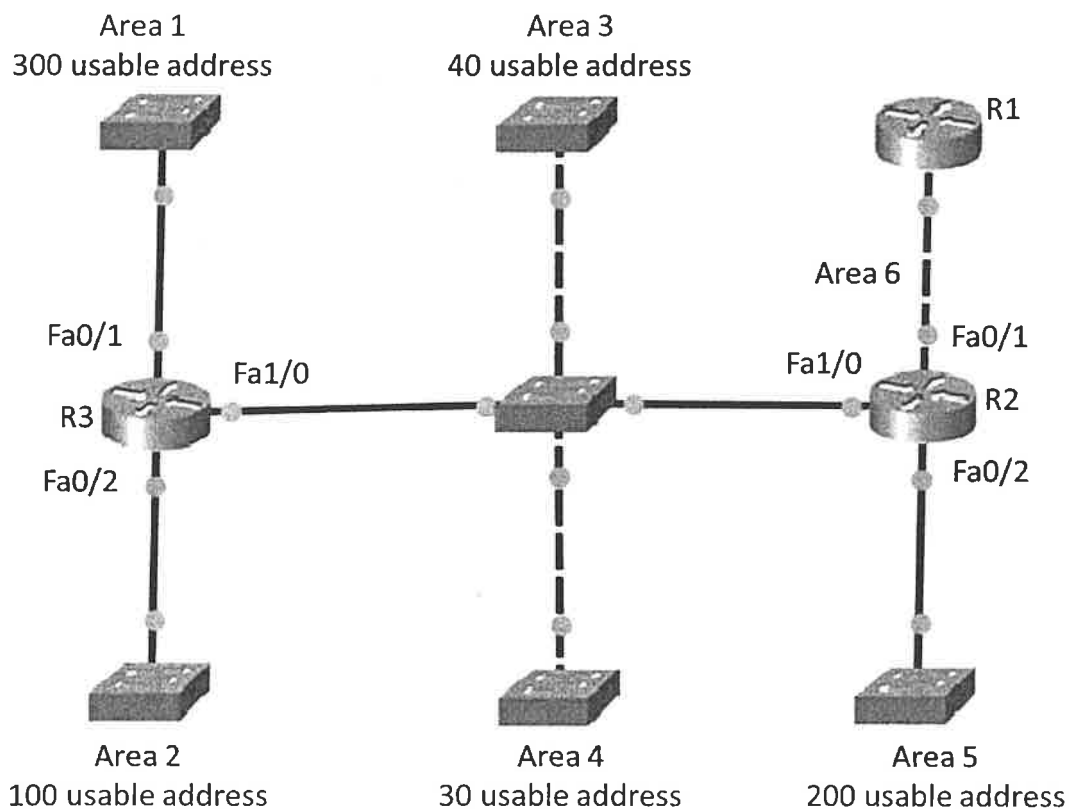


Figure 1

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Q2. Refer to the network in Figure 2. Company AA has a public IP address of 200.1.1.1. Network Address Translation (NAT) is configured on routers R1 to enable Internet access of Company AA server and hosts using the public IP address.

- (a) Due to security reasons, only hosts from the HR department with the IP address 192.168.1.1 to 192.168.1.31 are allowed to access the Internet. **Show** the relevant ACL and Port Address Translation configurations of router R1. Assume the necessary IP addresses have been configured in all the router interfaces. (Hint: You will only need to utilize the information given in Figure 2 for your setup). (15 marks)
 - (b) Supposed only the HTTPS server is allowed access to the Internet, while the DNS server is used within Company AA's network. Continue from Q2.(a), **show** the relevant Port Address Translation configurations on router R1. (6 marks)
 - (c) Continue from Q2.(b), **show** the relevant Port Forwarding configurations of router R1 to enable traffic from the Internet to access the HTTPS service. (4 marks)
- [Total : 25 marks]

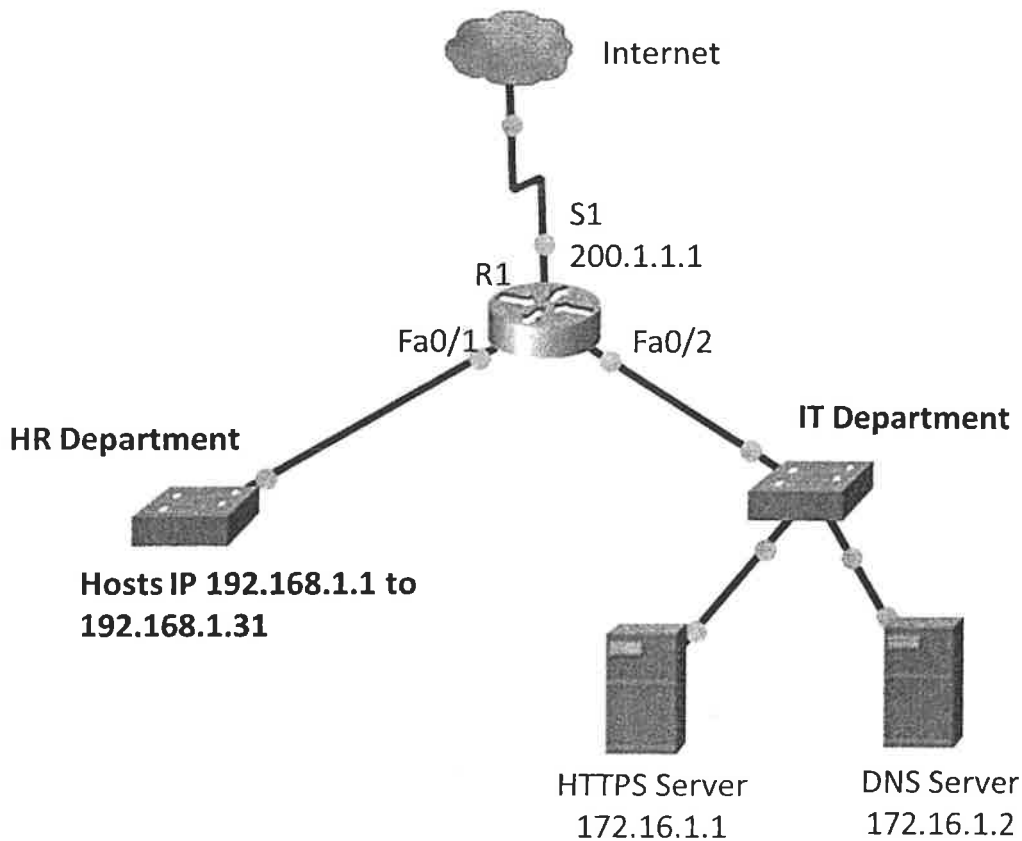
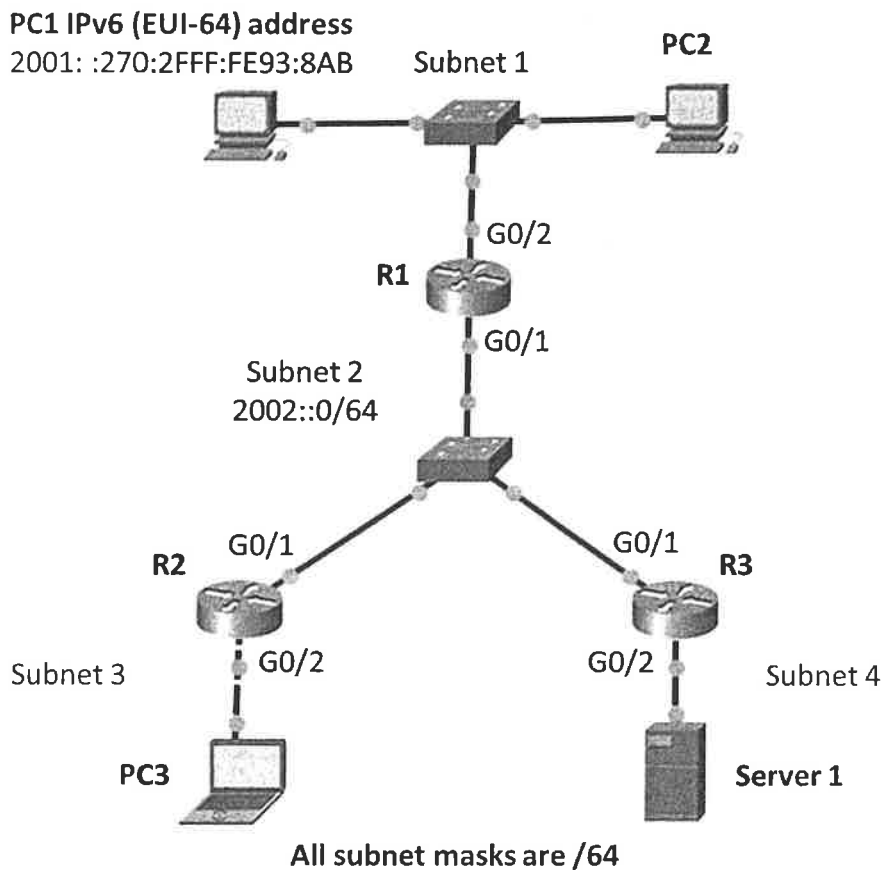


Figure 2

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- Q3. (a) An IPv6 only network is shown in Figure 3. Answer the following questions:
- (i) **Determine** the MAC address for PC1. (3 marks)
 - (ii) **Show** all the necessary commands to configure the gateway of Subnet 1 with the second-last usable address. (4 marks)
 - (iii) **Determine** the IPv6 interface address of R1, R2 and R3 in Subnet 2 if they are assigned the first usable, second usable and third usable address respectively. (3 marks)
 - (iv) **Determine** the network address of Subnet 3 and Subnet 4 if they are assigned with the next available subnet from Subnet 2. (4 marks)
 - (v) **Show** the static route configurations in R2 to enable PC3 to communicate with all other devices. (5 marks)
- (b) Briefly **discuss** THREE (3) techniques used for IPv4-IPv6 transition and coexistence. (6 marks)
- [Total : 25 marks]

**Figure 3**

UCCN2243 INTERNETWORKING PRINCIPLES AND PRACTICES**Section B (Choose Any One Question)**

- Q4. (a) A PC has been configured with the following iptables commands as shown in Table 1. **Explain** the actions of these commands on the specified network traffic to the PC. (6 marks)

```
#iptables -A INPUT -p icmp --icmp-type echo-request -m limit --limit 3/second -j ACCEPT

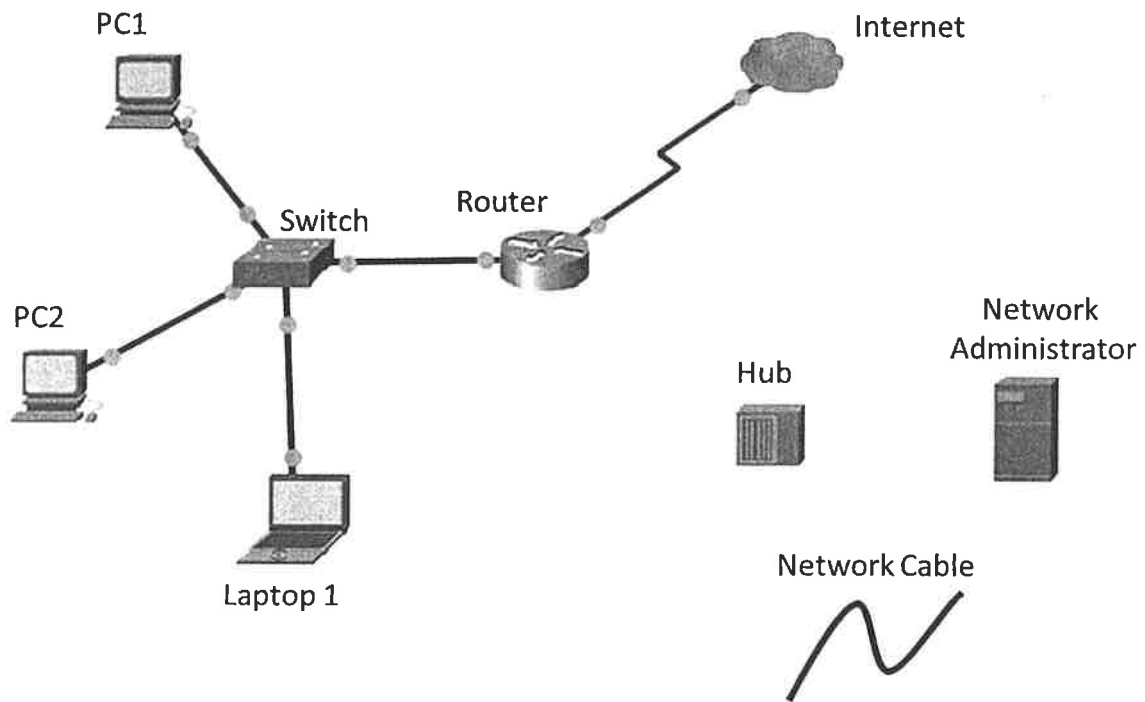
#iptables -A INPUT -p icmp --icmp-type echo-request -m limit --limit 5/minute -j LOG
```

Table 1

- (b) Given the network shown in Figure 4.
- (i) Utilizing the additional network components provided (i.e. Hub, Network Administrator and Network Cable), **redraw** the network to divert the Internet traffic of PC1 and PC2 towards the Network Administrator by using hubbing out. (7 marks)
 - (ii) Supposed PC2 installs Wireshark and observes its network traffic. **Discuss** how PC2 could determine if it is connected through a hub. (4 marks)
 - (iii) Apart from hubbing out, the ARP Spoofing is another method to capture network traffic from a target device on a switched network. **Explain** how the ARP is vulnerable to such exploitation. (6 marks)
 - (iv) **List TWO (2)** other methods to capture traffic from a target device on a switched network (2 marks)

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Q4. (Continued)

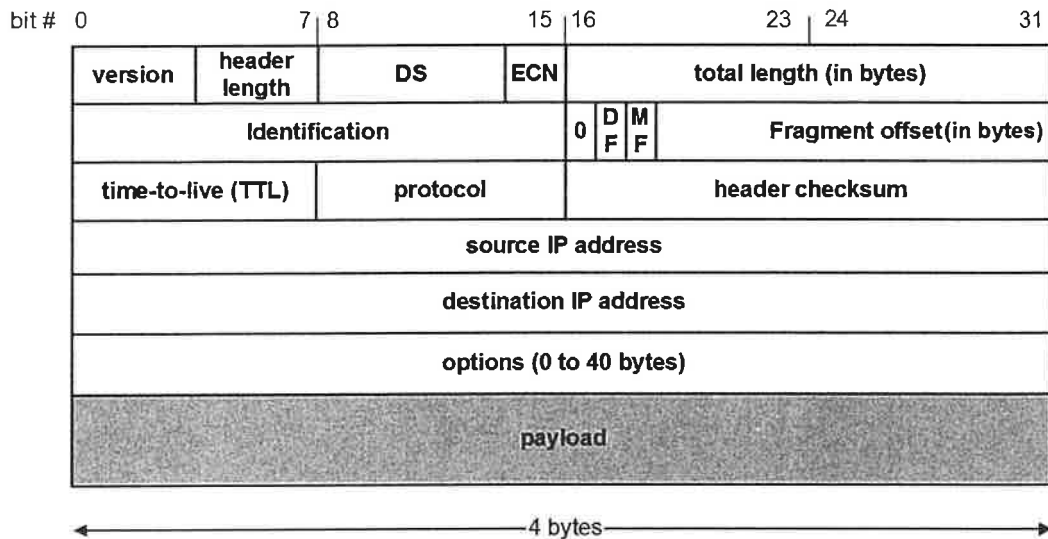
**Figure 4**

[Total : 25 marks]

- Q5. (a) Figure 5 shows the data dump (in hexadecimal) at the beginning of an IP packet. Answer the following:
- (i) **Calculate** the size (in bytes) of the IP data/payload. (4 marks)
 - (ii) **Determine** the size (in bytes) of the option field. (2 marks)
 - (iii) Continue from Q5 a(ii), suppose the datagram will be routed to an Ethernet network **calculate** the total length for each of the fragments (6 marks)
 - (iv) **Determine** the fragment offset for each of the fragments. (3 marks)

UCCN2243 INTERNETWORKING PRINCIPLES AND PRACTICES**Q5. (Continued)**

47 00 0F A4 10 00 00 00

**Figure 5**

- (b) **Discuss** THREE (3) disadvantages of fragmentation to the network. (6 marks)
- (c) Connection-oriented service is among the options provided by the Transport layer. **Describe** FOUR (4) characteristics of this connection. (4 marks)
[Total : 25 marks]

